

Sales Insights Dashboard – AdventureWorks

■ Project Overview

This project analyzes AdventureWorks sales data using SQL and Power BI. The goal is to provide insights into revenue, product performance, customer behavior, and regional trends. The project simulates a real-world business use case where management needs KPIs and dashboards to track performance.

■■ Tools & Technologies

- SQL Server – for data cleaning, transformations, and KPI queries
- Power BI – for building interactive dashboards and DAX measures
- GitHub – for project documentation and code sharing

■■ SQL Work

- Cleaned and prepared raw sales data by handling nulls and joining fact & dimension tables
- Created KPIs such as Total Sales, Profit Margin, YoY Growth, and Monthly Trends
- Wrote optimized queries to extract regional and product-level sales insights

■ Power BI Work

- Built an interactive dashboard with slicers for Year, Product, and Region
- Designed visuals: Bar charts for product sales, Maps for regional sales, Line charts for monthly trends, and KPI cards for totals
- Used DAX measures for dynamic calculations such as YoY Growth and Profit %

■ Key Insights

- A small set of top products contribute to 60% of overall sales
- Certain regions consistently underperform compared to others
- YoY growth analysis highlights strong and weak sales years
- Profit margins vary significantly across product categories

■ How to Run the Project

- 1. Restore AdventureWorks database in SQL Server
- 2. Run SQL scripts (data_cleaning.sql, sales_queries.sql, kpi_calculations.sql)
- 3. Open Sales_Insights.pbix in Power BI
- 4. Connect Power BI to SQL Server for live data and refresh

■ Conclusion

This project demonstrates end-to-end data analysis skills: working with SQL for data preparation, Power BI for visualization and storytelling, and GitHub for sharing and collaboration. It reflects real-world Data Analyst responsibilities in transforming raw data into actionable insights.